

BRIEF REPORTS

Exploring Developmental Differences in Visual Short-Term Memory and Working Memory

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Although visuospatial short-term memory tasks have been found to engage more executive resources than do their phonological counterparts, it remains unclear whether this is due to intrinsic differences between the tasks or differences in participants' experience with them. The authors found 11-year-olds' performances on both visual short-term and working memory tasks to be more greatly impaired by an executive suppression task (random number generation) than were those of 8-year-olds. Similar findings with adults (e.g., Kane & Engle, 2000) suggest that the imposition of a suppression task may have overloaded the older children's executive resources, which would otherwise be used for deploying strategies for performing the primary tasks. Conversely, the younger children, who probably never had the capacity or know-how to engage these facilitative strategies in the first place, performed more poorly in the single task condition but were less affected in the dual task condition. These findings suggest that differences in the children's ability to deploy task-relevant strategy are likely to account for at least part of the executive resource requirements of visual memory tasks.

Keywords: visuospatial working memory, executive functions, developmental differences, dual task interference, cognitive and mnemonic strategies

According to Baddeley and Hitch (1974), working memory is a system that provides temporary storage and simultaneous processing or manipulation of information. It has been shown to predict higher cognitive functions or intellectual abilities, such as mathematical skills (Bull & Scerif, 2001; Lee, Ng, & Ng, 2009; Lee, Ng, Ng, & Lim, 2004; Logie, Gilhooly, & Wynn, 1994), reading skills (Hitch, Towse, & Hutton, 2001), and intelligence (Engle, Tuholski, Laughlin, & Conway, 1999). Much of the earlier research focused on adults and the verbal component of the model. More recently, there has been interest in the visuospatial component

(e.g., Pickering, 2001b; Logie, 1995) and the developmental relevance of the model (e.g., Baddeley & Hitch, 2000).

The Visuospatial Sketchpad

Although there is general agreement that the visuospatial sketchpad is not a unitary system, there are differing views on the properties of the subcomponents (Logie, 1995; Rudkin, Pearson, & Logie, 2007). Logie (1995) proposed that the visuospatial component consists of a visual cache, which passively stores visual information, and an inner scribe, which actively rehearses spatial and sequential movement as well as visual information. Other researchers have proposed a static and dynamic separation (Pickering, 2001a; Pickering, Gathercole, Hall, & Lloyd, 2001). The distinction between the visual/static versus spatial/dynamic components is best characterized by the Visual Patterns Test and the Corsi blocks task. The Visual Patterns Test (Della Sala, Gray, Baddeley, & Wilson, 1997) involves the presentation and recall of static or simultaneously presented matrices. In contrast, the Corsi blocks task (Milner, 1971) involves the recall of movement sequences formed when an experimenter taps on a subset of blocks that have been randomly arranged among other blocks. In support of the visual/spatial divide, Della Sala, Gray, Baddeley, Allamano, and Wilson (1999) found that a visual interference task caused a decrease in performance on the Visual Patterns Test but not on the Corsi blocks task. Spatial tapping had the opposite effect.

The Present Study

Most previous research (e.g., Ang & Lee, 2008; Fisk & Sharp, 2003; Klauer & Stegmaier, 1997; Vandierendonck, Kemps, Fas-

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tame, & Szmalec, 2004) has focused on the relationship between the spatial subcomponent of the visuospatial sketchpad and the central executive. For the spatial subcomponent, at least, the distinction between short-term and working memory seems to be less clear than was proposed in the original working memory model (Baddeley & Hitch, 1974). We believe that the separation between visual short-term and working memory is also less distinct than originally proposed. It has been suggested that maintenance of visual information is effortful and demands executive resources (Baddeley, Cocchini, Della Sala, Logie, & Spinnler, 1999; Gathercole & Pickering, 2000). In a dual-task study, Rudkin et al. (2007) found that random number generation (RNG) impaired performance on a visual short-term memory task. This indicated the possibility of executive involvement in visual short-term memory tasks.

There are few developmental studies available that have examined the executive demands of visual memory tasks. In a study conducted with adults and children, Hamilton, Coates, and Hefernan (2003) examined the extent to which a verbal fluency task impaired performance on a visual short-term memory task when they were performed together. In the visual short-term memory task, participants were asked to decide whether a cartoon figure was identical to one shown earlier. Hamilton et al. found that although the adults performed better than the children on the visual span task alone, the verbal fluency task affected the adults' performance more than it did the children's. The implications of their findings were, however, subject to methodological complications. First, the performance of the verbal fluency task on its own was not measured; therefore, it is not known whether there was a trade-off in performance between the interference and visual tasks. Although the stimulus was more child friendly than the more commonly used visual matrices, body parts on the cartoon figure may have served as a verbal aid in the recall of to-be-remembered locations. A further problem was identified by Hamilton et al.: The verbal fluency interference task might not be suitable for children, as they might be limited in their vocabulary. Furthermore, the effects of the interference tasks may have been attenuated, because they were carried out not concurrently but during a 5-s maintenance interval between stimuli presentation and response. Nonetheless, this study raised the possibility that there may be age differences in the extent to which executive resources are recruited in performing visual short-term memory tasks.

Though it is known that performance on visual memory tasks improves with age, the mechanism that underpins this change is not yet fully understood (e.g., Gathercole, 1999; Logie & Pearson, 1997; Pickering, 2001b; Pickering et al., 2001; Swanson, 1996). There may also be developmental differences in the executive demands of visual short-term versus working memory tasks. In the adult context, Engle et al. (1999) suggested that a short-term memory task for some adult participants might be essentially a working memory task for others. Knowledge or experience and familiarity with a task can determine its executive demands; the less experience one has with a task, the more likely that executive resources will be drawn upon to complete the task. For tasks requiring visual memory, experience with visual stimuli will likely improve performance, and such improvement is likely to increase with age in children. This added experience may allow older children to easily "chunk" various parts of abstract patterns in such tasks into meaningful shapes or familiar images (Pickering,

2001b). Conversely, younger children may need to engage more executive resources in order to perform a visual short-term memory task, and this would make such tasks more like a working memory task in terms of executive demands.

However, in studies in which adults with low working memory span were compared with those with high working memory span, it has been found that though the high-span adults performed better than the low-span adults in single task conditions, they tended to be more impaired in dual task conditions (Kane & Engle, 2000; Rosen & Engle, 1997). High-span and low-span adults can be considered as corresponding to older and younger children respectively, as older children tend to have higher spans than do younger children. Rosen and Engle (1997) proposed that the working memory capacity of high-span adults was used to monitor and suppress previously retrieved category examples and to generate new examples in the verbal fluency task. In contrast, low-span adults had sufficient working memory capacity only for monitoring responses. When a concurrent task is imposed, it diverts attention from the verbal fluency task in high-span adults and causes a decrease in performance. The low-span adults were not affected because there was little attention that could be diverted from generating new responses in the first place. Kane and Engle (2000) found that high-span adults were less susceptible than low-span adults to proactive interference in a single task condition but became as susceptible as low-span adults in a dual task condition. They argued that high-span adults used attention to resist interference at encoding and retrieval whereas low-span adults did not; hence, high-span adults became more susceptible to proactive interference when their attention was divided (see also Gilhooly, Logie, & Wynn, 1999).

Our aim in the current study was to investigate whether there are differences in the recruitment of executive resources by 8- and 11-year-olds in performing visual short-term versus working memory tasks. Using a dual-task paradigm, we asked children to perform a RNG task together with a visual short-term memory task or a visual working memory task. In the RNG or executive suppression task, participants have to produce random number sequences and actively monitor their responses to avoid automatically producing familiar sequences (Baddeley, 1996; Miyake, Friedman, Emerson, Witzki, & Howerter, 2000; Towse & Neil, 1998). The task was selected because it places strong demands on executive resources and is not affected by variability in children's vocabulary skills.

We expected the older children to perform better than the younger children on both visual short-term and working memory tasks. Performance of the younger children on the visual memory tasks in the dual task condition was also expected to be more impaired than that of the older children. For the younger children, both the short-term and working memory tasks were expected to be equally impaired by the executive suppression task. For the older children, the working memory task was expected to be more demanding of executive resources than was the short-term memory task.

Method

Participants and Design

The participants were 106 children recruited from two primary schools and two student care centers in the western and eastern

parts of Singapore (sixty-one 8-year-olds: 28 boys, mean age = 8.20 years, range = 7.38–8.84 years, $SD = 4.36$ months; forty-five 11-year-olds: 24 boys, mean age = 11.09 years, range = 10.43–11.74 years, $SD = 4.47$ months). Participants were mainly of Chinese ethnic origin, with a few participants of Indian and Malay origin. English is the language of instruction at school and is the lingua franca in Singapore. With mother tongue taught at school and spoken at home by about half of the population, the majority of school-attending children are functionally bilingual. The children were mainly from lower to middle-income households. Further details regarding parental occupation, education, and ethnicity were not noted. In the study, we used a 2 (age: 8-year-olds vs. 11-year-olds) $\times$ 2 (task condition: single vs. dual task conditions) $\times$ 2 (task type: Visual Patterns Test vs. Patterns Memory task) mixed design. Power analyses showed that with alpha set at .05, a sample of this size has 85% power to detect a small difference with Cohen's $f = .16$.

Materials and Procedure

Visual short-term memory task. Visual short-term memory span was measured with the Visual Patterns Test (Della Sala et al., 1997), primarily because it was deemed a suitable task for testing children. This task has no obvious executive or processing requirement. The children had to remember simultaneously presented patterns made by blackened squares in grids of varying shapes and sizes. In each grid, half the squares were blackened. Figure 1 shows an example of a question from the task. The task increased in difficulty as the size of the grids increased from four to 30 squares. There were three questions for each grid size or difficulty level and 42 questions in total. There was one practice question with a grid of eight squares.

Each pattern was flashed for 3 s on a computer screen, after which the pattern had to be reproduced on a blank grid on the children's answer sheets. There was no time limit for responses. The experimenter made sure that each child understood the task and answered the practice question correctly before the actual task commenced. The task was terminated when the children could not recall correctly any of the three patterns at a particular difficulty level. Each correctly recalled pattern scored one point, and the maximum score was 42.

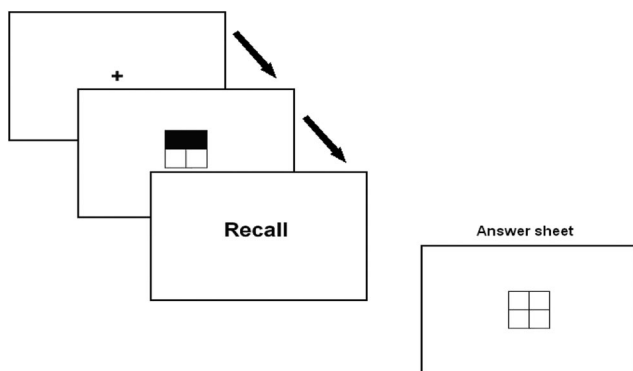


Figure 1. Schematic representation of a question from the Visual Patterns Test.

Visual working memory task. We devised the Patterns Memory task to measure visual working memory. This complex span task was adapted from tasks used in studies by Law, Morrin, and Pellegrino (1995) and Miyake, Friedman, Rettinger, Shah, and Hegarty (2001). A pilot study showed that this new task was suitable for testing children. Verification of a matrix equation made up of dots connected by lines was the processing component of this task, and the storage component required the children to remember locations of two blackened squares in a 4×4 grid. They were told to remember the grid and squares as a pattern. Each trial consisted of matrix equation–grid pairs in which a matrix equation was presented followed by a grid with blackened squares. The number of trials per question or span ranged from two to five, and there were five questions for each span and two practice questions of span two. Figure 2 illustrates an example of a question with a span of two.

The children had 8 s to verify the matrix equations (i.e., true or false) using the computer keyboard. The grid with two blackened squares was shown immediately after they responded. After all the required number of matrix equation–grid pairs were shown, a screen with the cue (i.e., one of the patterns shown previously but with only one blackened square) appeared. Only one pattern contained the cue. The children then had to draw on the answer grid the pattern that contained the cue (i.e., to shade in the second blackened square). There was no time limit to respond, and the children pressed the space bar to advance to the next question when they finished answering. Every correct pattern drawn scored one point, and the maximum score was 20. The verification component was not scored, but the children were not aware of this. An E-prime version of this task can be found at <http://acdl.crpp.nie.edu.sg/research/resources.htm>

Executive suppression task. The RNG task served as the concurrent executive suppression task. Though the task was complex and demanding, pilot testing showed it was suitable for use with children. The children were able to produce strings of random numbers and did not generate excessive strings of repeated or adjacent numbers. As previous studies with adults and children have shown that performance decreases when more response alternatives are used (Towse, 1998; Towse & McLachlan, 1999), only 10 response alternatives (i.e., numbers 1–10) were used.

The children performed this secondary task throughout the duration of the short-term and working memory tasks. They were asked to say aloud a number whenever they heard a tone (every 3 s, a rate that was optimal and minimized missing responses). A baseline measurement of the RNG task in which the children were asked to give 100 responses was taken before the dual task conditions were carried out. The experimenter explained to the children what the word *random* meant, and they were given a short practice session of about 10 responses. In the dual task conditions, no limit on the number of responses was set: The children continued giving responses until they finished the primary tasks. When they were not able to give a response for four to five consecutive tones, the experimenter reminded them.

A measure of performance on the RNG task was derived from the combined adjacency index from the RgCalc program (Towse & Neil, 1998). It is a percentage measure of the number of adjacent number pairs (e.g., 1 then 2, or 8 then 7) in the children's responses. Lower numbers denote better performance. Only numbers from 1 to 10 were considered legitimate responses and were

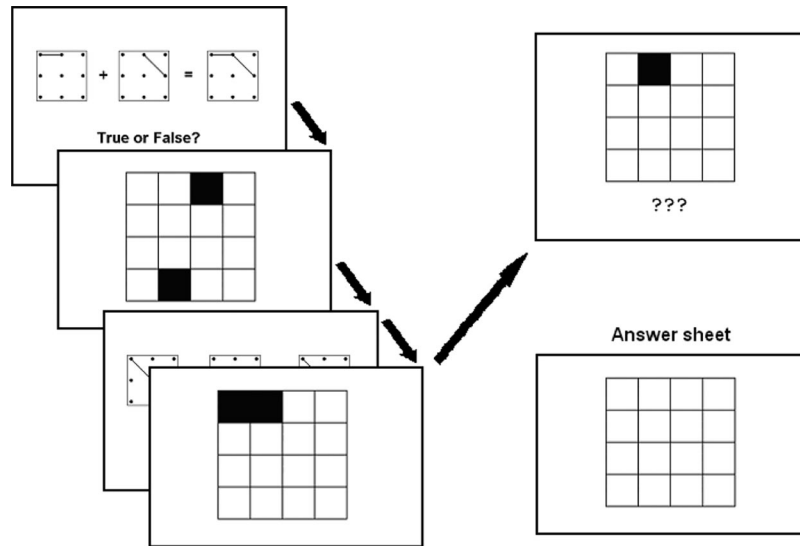


Figure 2. Schematic representation of a question from the Patterns Memory task.

included in the calculation of the adjacency index. Adjacency scores have been shown to differ in different task conditions, such as between single and dual task conditions (Towse & Valentine, 1997), so the index was deemed a suitable measure.

Both visual tasks were computerized using the E-prime software and were presented on laptops with 14.1-in. screens. The children were tested individually in quiet rooms in two to three sessions as part of the data collection for a larger study. Each session lasted 40 min to 1 hr. Due to time constraints imposed by the schools, one group carried out the tasks in two to three sessions over a number of weeks, and the other groups completed the tasks within a few days. Each task was first administered on its own, in the single task condition. This was followed by the dual task condition in which each task was paired with the RNG task. This sequence was not counterbalanced, because the dual task condition was known to be more difficult. Starting with the dual task condition might have discouraged participants and lowered their confidence and motivation. Order of presentation of the types of task was counterbalanced, and tasks were carried out in a number of predetermined sequences.

Results

The percentage accuracy scores from each task were used in the analyses to enable comparison, as the tasks originally had different

scales. The data were screened for outliers, and none were found. Skewness and kurtosis values were also examined, and all were approximately normal.

Table 1 shows the means and standard deviations for each visual task. A three-way mixed design analysis of variance (ANOVA) was performed. There was a significant interaction effect between task condition and age, $F(1, 104) = 8.36, p < .01, \eta_p^2 = .07$. The older children performed better on both visual tasks in both task conditions, but their performance in the dual task condition, $t(44) = 7.49, p < .001, \eta^2 = .57$, was impaired to a larger extent than was the younger children's performance, $t(60) = 6.12, p < .001, \eta^2 = .39$. Figure 3 illustrates this relationship. The interaction between task type and task condition was also significant, $F(1, 104) = 31.07, p < .001, \eta_p^2 = .23$ (see Figure 4). Mean percentage accuracy scores on both tasks were similar in the single task condition and were higher than those in the dual task condition. Performance on the Patterns Memory task was impaired to a greater extent by the RNG task than by the Visual Patterns Test. Bonferroni-corrected t tests showed that performances on both the Visual Patterns Test and the Patterns Memory task were significantly impaired in the dual task condition, $t(105) = 4.80, \eta^2 = .18$, and $t(105) = 9.07, \eta^2 = .44$, respectively ($p < .001$). The tertiary interaction and the interaction between task type and age were not significant.

Table 1
Mean and Standard Deviation of Percentage Accuracy Scores of Visual Short-Term Memory and Working Memory Tasks

Task	8-year-olds ($n = 61$)		11-year-olds ($n = 45$)	
	Single	Dual	Single	Dual
Visual Patterns Test	38.25 (8.73)	36.10 (9.67)	49.68 (10.61)	41.43 (14.24)
Patterns Memory task	37.30 (15.32)	25.41 (13.91)	48.78 (16.66)	31.56 (19.51)

Note. Data are for single and dual task conditions. Values enclosed in parentheses represent standard deviations.

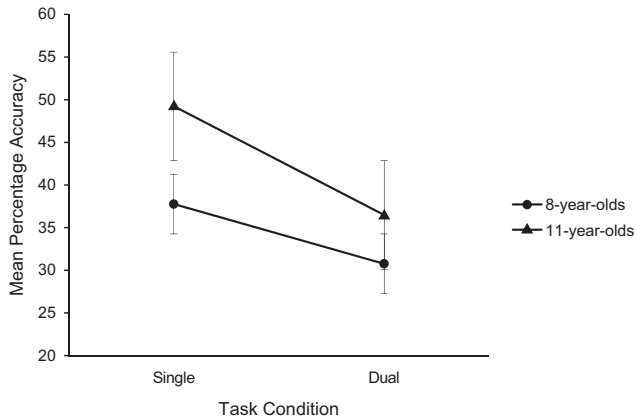


Figure 3. Mean percentage accuracy scores showing the interaction between task condition and age. The scores refer to averaged performance scores across the Visual Patterns Test and the Patterns Memory task. The error bars show standard errors of the means.

Analysis of RNG Performance

The adjacency scores were analyzed to examine differences in the performances of the RNG task when it was administered on its own versus when it was administered with the visual tasks. The data were screened for outliers. One data point from the Visual Patterns Test data was an outlier with a value greater than 3 standard deviations from the mean and was replaced by the value at 3 standard deviations from the mean. Examination of the skewness value after replacement of the outlier showed that its distribution was approximately normal. Table 2 shows the means and standard deviations.

An ANOVA based on a 2 (age: 8-year-olds vs. 11-year-olds) $\times$ 3 (task situation: RNG single, RNG with Visual Patterns Test, RNG with Patterns Memory task) mixed design was used. The main effect of task situation was marginally significant, $F(2, 103) = 3.02$, $p = .053$, $\eta_p^2 = .06$. All the children tended to produce fewer adjacent pairs of numbers in the single task situa-

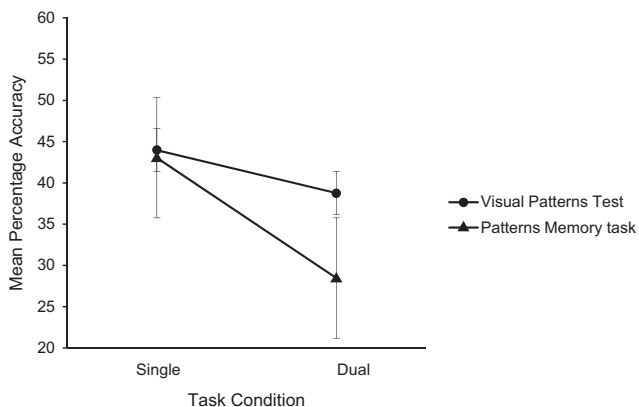


Figure 4. Mean percentage accuracy scores showing the interaction between task condition and task type. The scores refer to averaged performance scores across both age-groups. The error bars show standard errors of the means.

Table 2

Mean and Standard Deviation of Random Number Generation Adjacency Scores

Task situation	8-year-olds ($n = 61$)	11-year-olds ($n = 45$)
Single	25.15 (9.90)	23.22 (9.16)
Dual: RNG with VPT	26.50 (9.79)	26.52 (11.48)
Dual: RNG with PM	27.44 (9.92)	26.23 (9.07)

Note. RNG = random number generation; VPT = Visual Patterns Test; PM = Patterns Memory task. Values enclosed in parentheses represent standard deviations. Higher scores indicate poorer performance.

tion than in the dual task situations. There were no other significant main or interaction effects.

Discussion

This study used a dual-task methodology in investigating the executive demands a visual short-term memory task and a visual working memory task placed on 8- and 11-year-olds. As predicted, the older children performed better than the younger children on both visual short-term memory and working memory tasks in both single and dual task conditions. However, contrary to expectations, the older children were more affected by the executive suppression task than were the younger children across both types of memory tasks. Analysis of the executive suppression task showed little evidence of a strategic trade-off in the dual task condition: The children did not seem to be sacrificing performance on the primary tasks for the sake of the executive suppression task. Our results showed that across both age-groups the working memory task was more impaired by the executive suppression task than was the short-term memory task. Notably, this latter finding was not moderated by age. We had expected the younger children's performance on the short-term memory task to be as badly affected by the suppression task as was that on the working memory task. Nonetheless, our present finding provides evidence that, in the visual domain, the working memory task places a greater demand on executive resources than does the short-term memory task.

Developmental Differences in Executive Demands of the Visual Short-Term and Working Memory Tasks

The finding that older children were more affected by the executive suppression task than were the younger children cannot be fully explained by the older children possessing more task-specific knowledge or experience. If being younger is a mark of possessing less experience, and less experience and proficiency in such memory tasks means that more effort or controlled attention is required for maintaining or manipulating a visual image, such mental efforts are likely to require greater executive control. However, the older children seemed to be engaging executive resources to a greater extent instead.

Our findings seem to be analogous to findings in previous studies by Engle and colleagues, with the older/younger children behaving similarly to the high- and low-span adults (Kane & Engle, 2000; Rosen & Engle, 1997). The older children were more detrimentally affected by the imposition of a secondary task than were the younger children. One interpretation is that the older

children, possessing a greater working memory capacity than the younger children, use these resources to deploy mnemonic strategies that enhance their performance on the primary visual tasks. When an executive suppression task is imposed, it overloads the older children's capacity and removes their ability to deploy mnemonic strategies. The suppression task may also directly impair primary task performance. The performance of the younger children, who do not possess the resources to enhance their task performance in the first place, is relatively preserved because the suppression task has only a direct effect—but not an indirect effect, via mnemonic strategies—on their performance on the primary task.

Anecdotal evidence from a related experiment using the same tasks showed that a number of the children, especially the older children, reported imagining lines joining the blackened squares in the patterns, naming the shapes formed when possible, and counting the blackened squares. Experimenters also observed children using their fingers to mark locations of the blackened squares on the answer sheet. Such strategies are likely to be demanding and effortful, as they require executive resources. The older children seemingly employed such strategies more extensively or used more elaborate strategies. There is a wealth of evidence showing that older children use more sophisticated mnemonic strategies in general than do younger children (DeMarie & Ferron, 2003).

We expected that the short-term memory task and the working memory task would be impaired to a similar extent for the younger children. Their lack of experience with such tasks was expected to lead them to engage more executive resources than would the older children for the short-term memory task. However, across both age-groups, performance on the working memory task was more impaired than performance on the short-term memory task by the executive suppression task. One reason for this finding could be a lack of statistical power. Though this explanation cannot be ruled out, the effect size would be very small, as power analyses have shown that the sample was large enough for us to detect even small differences. It is possible that our original assumption that experience is a mediator was inaccurate. There may not be a significant difference in task experience and proficiency between the 8- and 11-year-olds. However, our data show that the older children were more accurate than the younger children on both tasks. One speculative explanation is that 8-year-olds, though not as proficient as 11-year-olds, are sufficiently proficient on the visual short-term memory task to not require a significant amount of executive resources to perform the task. Nonetheless, they were unable to deploy the more sophisticated mnemonics as used by the older children. The use of mnemonics resulted in both age differences in task performance in the nonsuppression condition and the older children's greater susceptibility to suppression effects.

Is Suppression Caused by Executive or Nonexecutive Demands?

One alternative explanation that the reported data are unable to rule out is that the decrease in task performance may have been caused simply by performing two tasks concurrently or by the presence of common nonexecutive task demands instead of common executive demands between the visual memory tasks and the executive suppression task. To rule out this explanation, we conducted a similar experiment with another group of children in

which the same visual tasks were paired with an articulatory suppression task. As articulatory suppression should draw only on phonological loop resources and differs from the executive suppression task in terms of the demands it places on the central executive, it was expected to have little or no effect on the visual tasks. The materials and procedures were unchanged from the first experiment.

Across both age-groups, when the executive suppression task was paired with the visual memory tasks, the working memory task was more affected, but when the articulatory suppression task was paired with the visual memory tasks, only the short-term memory task was affected. Across the two experiments, the executive suppression task had a greater detrimental effect on the working memory task than did articulatory suppression. These findings suggest that the effect of the executive suppression task is caused by executive rather than nonexecutive interference.

Caveat

An important caveat to note is the separation between short-term and working memory tasks and the distinction between task versus construct measurement. If traditional measures of visuospatial short-term memory, such as the Visual Patterns Test, are impaired by an executive suppression task, these tasks are often assumed to have an executive component. However, such findings also raise the issue of whether these tasks are actually measuring short-term memory or working memory. Unfortunately, there are few tasks that are considered "pure" measures of visuospatial short-term memory (Pickering, 2001b); the Visual Patterns Test is widely regarded and used as such a measure of the visual short-term memory construct. Further, we found that across both age-groups, the working memory task placed greater demands on executive resources than did the short-term memory task. This finding provides evidence that the Visual Patterns Test was indeed measuring short-term memory rather than working memory, as did the Patterns Memory task.

Conclusion

Our study shows that though the older children may have more knowledge or are more familiar with the demands of visual memory tasks, they seem to rely more heavily on executive resources when performing these short-term and working memory tasks. This resulted in greater impairment by the executive suppression task for the older children than for the younger children in the dual task condition, even though their task performance was better in the single task condition. Our findings can be seen as an extension of those of Engle and colleagues to children's performance on visual memory tasks, in which the better performance of the older children, like that of high-span adults, was due to the engagement of executive resources (Kane & Engle, 2000; Rosen & Engle, 1997). We reasoned that this greater impairment was because the older children employed effortful mnemonic strategies to enhance their task performance and also engaged executive resources to cope with normal task demands. When the executive suppression task was imposed, it prevented the use of such strategies and directly decreased performance on the primary tasks. The younger children, who were not using such strategies in the first place, were less affected by the suppression task. Because we

did not measure strategy use directly, further research should focus on such direct measurement and on the type of strategies children use to enhance task performance. Other future studies could involve training of children on the use of mnemonic strategies or giving instructions that emphasize the use of such strategies.

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